

RECEIVING REPORT

Receiver No.: 18556227

Document Date: 05/27/2025

Posting Date: 05/27/2025

Expiration Date:

P.O.#: 5300130329

Item #: 3

Batch#: ACE0138413

Insp. Lot: 000001319866

Plant: ACE

Plant Description: Clearfield-Solumina

Vendor Name: HEXCEL CORP

Vendor Addr: 1913 N KING ST

SEGUIN

TX 78155

Quality Inspection

Unloading Point:

Vendor No.: 90069901

Buyer Name: Keith Larsen

SQAR Code:

Project ID: ASBUX

Test Lab:

WBS:

Inspection Lot/Type:

Rev No.: 01

Material: MSRR4040-VAFR

Received UOM: LB

Base UOM: LB

Base UOM Received Qty: 337.780

Vendor Mtl. Number:

Mtl. Description: SATIN WEAVE CARBON FIBRE REINFORCED BISM

Mtl. Specification:

Alt. Material #:

Matl. Basic Data Text:

MSRR4040-VAFR:SATIN WEAVE CARBON FIBRE REINFORCED BISM

Bill of Lading: 92SPS055968

Receipt Created by: N15071

Head text:

Delivery Note: 92SPS055968

Received Qty: 337.780

Receipt Delivered To: _____

WAREHOUSE INFORMATION



Plt: ACE



Part Number: MSRR4040-VAFR



Qty: 337.78



Receiver No: 18556227

Storage Bin:



Order Number:

Warehouse: LIN1



Stock Type: Q



WBS Element: A-46Z01-NW.N46Z01000714



Inventory WBS: Y-ROLLS

Bin:

Storage Location: ACED

Item Text:

SATIN WEAVE CARBON FIBRE REINFORCED BISMALEIMIDE PREIMPREGN
ATTENTION: The revision level for this part is: 4
NG PROJECT: 46001.IN.INV.0RAW, PRIME CONTRACT:
Ship via: Best Way.

Qty. Insp.	Insp. Stamp No.	By Inspector	Qty. Rej.
Rej. No.	Stocked By	Date Stkd.	Stk. Loc.



Packing list

Document 92SPS055968
Date 27-May-2025
Printed date 27-May-2025

Customer P.O.



Customer P.O.
Account

5300130329
C000079

Packing Slip ID



Hexcel order
Account

92SO024459
C000079

Invoice address

Northrop Grumman Systems Corporation
Enterprise Transaction Services
8710 Freeport Parkway, Suite 200
Irving, TX 75063
United States

Delivery address

Northrop Grumman Systems Corporation
1051 S. Indust. Parkway
(ACCE)
Clearfield, UT 84015
United States

18556225
18556227

Delivery contact

Keith Larsen

Line	Item	Description		Selling qty		
5.1	111545	HexPly® F3C 433/M65;38%;285AW;50"		4,031.78 lb		
3						
Physical dimensions		W = 50.00				
Customer part		MSRR4040-VAFR				
Specification/Issue		MSRR4040/Rev ISSUE 4				
Ship date		27-May-2025				
Sales line LotId		9216633693				
Trade compliance						
Country of origin		US	Commodity code	6815130000 / Other articles of carbon fiber		
ECCN		1C010	End user country	US		
These items are controlled by the U.S. Government and authorized for export only to the country of ultimate destination for use by the ultimate consignee or end-user(s) herein identified. They may not be resold, transferred, or otherwise disposed of, to any other country or to any person other than the authorized ultimate consignee or end-user(s), either in their original form or after being incorporated into other items, without first obtaining approval from the U.S. government or as otherwise authorized by U.S. law and regulations.						
Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0103015	92M0103015001	65.91lb	65.91lb	65.91lb
		92M0103015	92M0103015002	72.19lb	72.18lb	72.18lb
		92M0103015	92M0103015003	70.62lb	70.61lb	70.61lb
		92M0103015	92M0103015004	81.60lb	81.60lb	81.60lb
		92M0103015	92M0103015005	89.05lb	89.05lb	89.05lb
		92M0103015	92M0103015006	85.13lb	85.13lb	85.13lb
		92M0103015	92M0103015007	85.52lb	85.52lb	85.52lb
		92M0103015	92M0103015008	60.42lb	60.41lb	60.41lb
		92M0103015	92M0103015009	12.55lb	12.55lb	12.55lb
		92M0103015	92M0103015010	87.09lb	87.09lb	87.09lb
		92M0103015	92M0103015011	83.95lb	83.95lb	83.95lb



Hexcel Corporation
6700 West 5400 South
West Valley City, UT 84118
United States



Packing list

Document 92SPS055968
Date 27-May-2025
Printed date 27-May-2025

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0103015	92M0103015012	78.46lb	78.46lb	78.46lb
		92M0103015	92M0103015013	96.12lb	96.11lb	96.11lb
		92M0103015	92M0103015014	39.23lb	39.23lb	39.23lb
		92M0103015	92M0103015015	9.42lb	9.42lb	9.42lb
		92M0103015	92M0103015016	82.39lb	82.38lb	82.38lb
		92M0103015	92M0103015017	30.99lb	30.99lb	30.99lb
		92M0103015	92M0103015018	88.27lb	88.27lb	88.27lb
		92M0103015	92M0103015019	81.99lb	81.99lb	81.99lb
		92M0103015	92M0103015020	87.49lb	87.48lb	87.48lb
		92M0103015	92M0103015021	40.41lb	40.41lb	40.41lb
		92M0103015	92M0103015022	45.51lb	45.51lb	45.51lb
		92M0103015	92M0103015023	75.32lb	75.32lb	75.32lb
		92M0103015	92M0103015024	77.29lb	77.28lb	77.28lb
		92M0103015	92M0103015025	15.69lb	15.69lb	15.69lb
		92M0103015	92M0103015026	117.30lb	117.30lb	117.30lb
		92M0103015	92M0103015027	57.28lb	57.28lb	57.28lb
		92M0103015	92M0103015028	77.29lb	77.28lb	77.28lb
		92M0103015	92M0103015029	85.13lb	85.13lb	85.13lb
		92M0103015	92M0103015030	88.27lb	88.27lb	88.27lb
		92M0103015	92M0103015031	87.09lb	87.09lb	87.09lb
		92M0103015	92M0103015032	112.59lb	112.59lb	112.59lb
		92M0103015	92M0103015033	58.45lb	58.45lb	58.45lb
		92M0103015	92M0103015034	89.45lb	89.44lb	89.44lb
		92M0103015	92M0103015035	85.52lb	85.52lb	85.52lb
		92M0103015	92M0103015036	81.21lb	81.21lb	81.21lb
		92M0103015	92M0103015037	89.45lb	89.44lb	89.44lb
		92M0103015	92M0103015038	87.49lb	87.48lb	87.48lb
		92M0103015	92M0103015039	86.70lb	86.70lb	86.70lb
		92M0103015	92M0103015040	87.88lb	87.88lb	87.88lb
		92M0103015	92M0103015041	85.13lb	85.13lb	85.13lb
		92M0103015	92M0103015042	88.66lb	88.66lb	88.66lb
		92M0103015	92M0103015043	85.92lb	85.91lb	85.91lb
		92M0103015	92M0103015044	54.92lb	54.92lb	54.92lb
		92M0103015	92M0103015045	95.72lb	95.72lb	95.72lb
		92M0103015	92M0103015046	88.27lb	88.27lb	88.27lb
		92M0103015	92M0103015047	87.88lb	87.88lb	87.88lb
		92M0103015	92M0103015048	87.88lb	87.88lb	87.88lb
		92M0103015	92M0103015049	85.92lb	85.91lb	85.91lb
		92M0105492	92M0105492001	83.95lb	83.95lb	83.95lb



Packing list

Document 92SPS055968
Date 27-May-2025
Printed date 27-May-2025

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0105492	92M0105492002	51.79lb	51.78lb	51.78lb
		92M0105492	92M0105492003	88.27lb	88.27lb	88.27lb
		92M0105492	92M0105492004	86.31lb	86.31lb	86.31lb
		92M0105492	92M0105492005	27.46lb	27.46lb	27.46lb
Total per Batch 92M0103015				3,694.00lb		
Total per Batch 92M0105492				337.78lb		
Total per Item 111545				4,031.78lb	4,031.65lb	4,031.65lb

Line	Item	Description	Selling qty
6 3	92TempRec-Ambient	92TempRec-Ambient	2.00 ea
Customer part TEMREC			
Ship date 27-May-2025			
Sales line LotId 9216633694			
Trade compliance			
Country of origin RS		Commodity code 9025198085 / TEMPERATURE RECORDER	
ECCN EAR99		End user country US	
These items are controlled by the U.S. Government and authorized for export only to the country of ultimate destination for use by the ultimate consignee or end-user(s) herein identified. They may not be resold, transferred, or otherwise disposed of, to any other country or to any person other than the authorized ultimate consignee or end-user(s), either in their original form or after being incorporated into other items, without first obtaining approval from the U.S. government or as otherwise authorized by U.S. law and regulations.			

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
				2.00ea	0.00lb	
Total per Batch				2.00ea		
Total per Item 92TempRec-Ambient				2.00ea	0.00lb	

Shipping Instructions				Carrier	Customer Pick-Up		
Mode of Delivery		Ground		Delivery Terms		FCA, Salt Lake City, UT, USA - Incoterms®2010	
Pallet IDs	Box IDs	Batches	Serial Nos			Net weight	Gross weight
0	0	2	54			4,031.65lb	4,031.65lb
EU TMS Package details							
Line 5.1 0							
Line 6 0							



21 May 2025

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5300130329 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285A; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0105492 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0105492 Date of Manufacture: 7 May 2025

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0103522 Manufactured by: Hexcel

Glass Lot No: 31W0027667 Manufactured by: Hexcel

- I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.
- II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Laney Defa 

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
 Configuration Number: CN1301
 Material Type: F3C 433/M65;38%;285AW;50"
 Lot Number: 92M0105492
 Order Number: 92M0105492

Report Printed: 17-May-2025 10:47:53AM
 Status: **PASS**
 Approval: **BOBADIEX**

1. Tensile, SBS, and TG Panel

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	Laminate Density By Immersion Avg (g/cc)		1.5100
	Laminate Density By Immersion Std (g/cc)		0.0013
	Laminate Density By Immersion CoV		0.09
	Dimensional Fiber Volume (%)		53.16 - 52.86 - 52.95
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.154
	Dimensional Fiber Volume CoV		0.29
	Void Volume (%)		0.8 - 0.9 - 0.9
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.047
	Void Volume CoV		5.41

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	SBS Strength (MPa)		84.5 - 80.9 - 80.9 - 73.9 - 81.5
	SBS Strength Avg (MPa)	MIN= 65	80
	SBS Strength Std (MPa)		3.87
	SBS Strength CoV		4.82

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	SBS Strength (MPa)		65.3 - 64.8 - 64.2 - 65.7 - 64.2
	SBS Strength Avg (MPa)	MIN= 50	65
	SBS Strength Std (MPa)		0.66
	SBS Strength CoV		1.02

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	Tg by DMA in °C (° C)		270.3
	Tg by DMA in °C Avg (° C)	MIN= 225.0	270.3



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0105492
Order Number: 92M0105492

Report Printed: 17-May-2025 10:47:53AM

Status: **PASS**
Approval: **BOBADIEX**

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		0.59

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0105492001/001	Norm Tensile Strength (MPa)		990.6 - 926.3 - 1001.7 - 985.2 - 974.8
	Norm Tensile Strength Avg (MPa)	MIN= 700	976
	Norm Tensile Strength Std (MPa)		29.27
	Norm Tensile Strength CoV		3.00
	Norm Modulus (GPa)		62.8 - 61.7 - 62.7 - 62.0 - 61.6
	Norm Modulus Avg (GPa)	MIN= 55	62
	Norm Modulus Std (GPa)		0.538
	Norm Modulus CoV		0.87

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0105492001	Volatile content @100°C in a Pan (%)		0.6 - 0.5
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1
92M0105492005	Volatile content @100°C in a Pan (%)		0.5 - 0.5
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Tack

Roll	Test/Result:	Requirements:	Values:
92M0105492001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0105492003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0105492
Order Number: 92M0105492

Report Printed: 17-May-2025 10:47:53AM

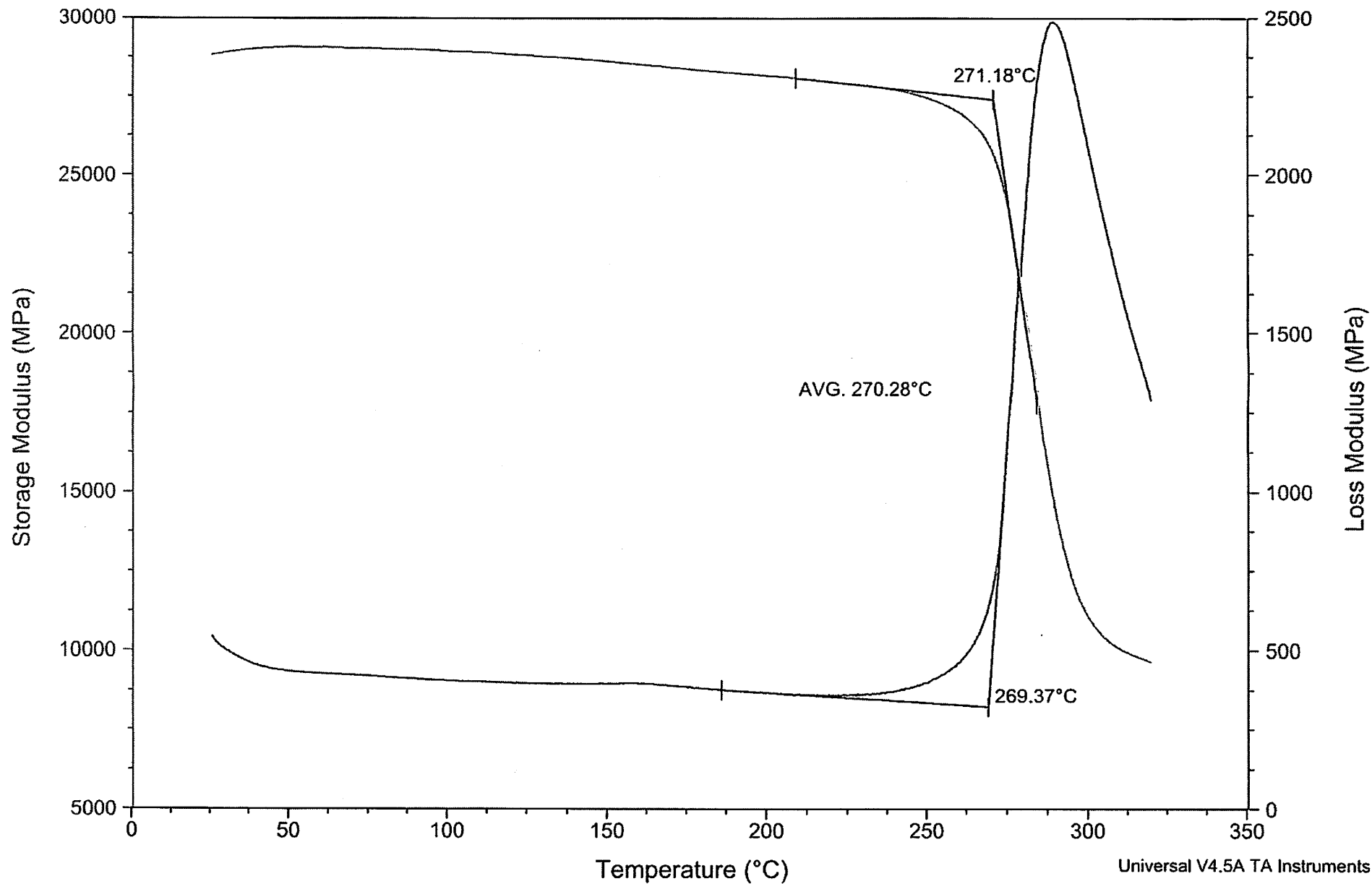
Status: **PASS**
Approval: **BOBADIEX**

____ End of report ____

Sample: 92M0105492-001 SP#001
Size: 34.8100 x 9.8400 x 2.3900 mm
Method: Temperature Ramp

DMA

File: \\...\\DMA Q800-2357\\8552\\92M0105492.001
Operator: EB
Run Date: 17-May-2025 08:05
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: 92M0105492

5/21/2025

Page 1 of 2

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0105492001	0	1	4.5904	2.8200	38.6	282	459.0			PETERSDX	05/07/2025
		2	4.6366	2.8338	38.9	283	463.7			PETERSDX	05/07/2025
		3	4.5989	2.8093	38.9	281	459.9			PETERSDX	05/07/2025
		Avg.	4.6086		39	282	461	50.30	A	PETERSDX	05/07/2025
92M 0105492002	0	1	4.5765	2.8219	38.3	282	457.6			PETERSDX	05/07/2025
		2	4.6118	2.8087	39.1	281	461.2			PETERSDX	05/07/2025
		3	4.5926	2.8185	38.6	282	459.3			PETERSDX	05/07/2025
		Avg.	4.5936		39	282	459	50.30	A	PETERSDX	05/07/2025
92M 0105492003	0	1	4.6022	2.8686	37.7	287	460.2			PETERSDX	05/07/2025
		2	4.5901	2.8684	37.5	287	459.0			PETERSDX	05/07/2025
		3	4.6151	2.8438	38.4	284	461.5			PETERSDX	05/07/2025
		Avg.	4.6025		38	286	460	50.30	A	PETERSDX	05/07/2025
92M 0105492004	0	1	4.5733	2.8487	37.7	285	457.3			PETERSDX	05/07/2025
		2	4.6016	2.8435	38.2	284	460.2			PETERSDX	05/07/2025
		3	4.5531	2.8462	37.5	285	455.3			PETERSDX	05/07/2025
		Avg.	4.5760		38	285	458	50.20	A	PETERSDX	05/07/2025
92M 0105492005	0	1	4.5486	2.8096	38.2	281	454.9			PETERSDX	05/07/2025
		2	4.5953	2.8377	38.2	284	459.5			PETERSDX	05/07/2025
		3	4.5779	2.8352	38.1	284	457.8			PETERSDX	05/07/2025
		Avg.	4.5739		38	283	457	50.30	A	PETERSDX	05/07/2025

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: 92M0105492

05/21/2025

Page 2 of 2

Spool	Retest:	RC	AWT	PAW	Stamp
92M0105492001	0	39	282	461	A
92M0105492002	0	39	282	459	A
92M0105492003	0	38	286	460	A
92M0105492004	0	38	285	458	A
92M0105492005	0	38	283	457	A
Averages:		38.3	283.4	459.0	



Product Certification

Page 1 of 4

COA Number:	SEG-COA-28205	Prod Order:	DOM:	Description:
Version:	1	31W0027667	24-Mar-2025	F3C433(FBL2)-50"
Issue Date:	11-Apr-2025	Weave Pattern:	5 Harness Satin	

Specifications
F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 5 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50inch - MSRR4040 Issue.4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result
31G0000367905	31W0027667	Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367906		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367907		Width	49.5 - 50.5 in	50.0
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367908		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367909		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.3

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 2 of 4

31G0000367909	31W0027667	Dried FAW Avg	278 - 292 g/m ²	283
31G0000367910		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367911		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367912		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367913		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367914		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367915		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367916		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367917		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5

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Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

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31G0000367917	31W0027667	Dried FAW Avg	278 - 292 g/m ²	284
31G0000367918		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367919		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367920		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367921		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367922		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000367923		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000367924		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367925		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5

This is to certify that material shipped against this order meets the specification(s) listed above.

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1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

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31G0000367925	31W0027667	Dried FAW Avg	278 - 292 g/m ²	283
31G0000367926		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367927		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367928		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367929		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367930		Width	49.5 - 50.5 in	50.3
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367931		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	282

Certification Exceptions/Additional Information

QMS07045, REV. 1, SEC. 5E

WARP: 240920210

FILL: 240920210

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Michael Galvan
Certification Technician



TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

Certificate of Analysis

QAP 004F2 Rev M

Customer: Hexcel Reinforcement Corp (TX) Contact: Patrick Splackermann;
Amy Shehane; Jerry Email/Fax: Patrick.Splackermann@hexcel.com;
Tschoepe; Greg Early amy.shehane@hexcel.com;
jerry.tschoepe@; Greg.Early@

CustomerPO#: 31PO0013988 RELEASE 1 Order: SH001381 Ship Date: 3/5/2025

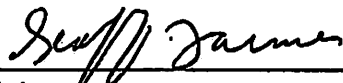
Customer P/N: n/a Product: Tenax™-E HTA40 E13 3K 200tex

Part Number: G305303 Quantity: 3000.000 kg Certifies To: MS-X-CTC-105 Rev B; MSRR9317/11
Issue 1

Lot Number: 240920210

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>cv%</u>
Tensile Strength, MPA (ksi):	4213.2 (611)	3.8
Tensile Modulus, GPA (msi):	238.1 (34.6)	0.9
Elongation, %:	1.67	3.7
Linear Density (without size), g/m:	0.2003	0.8
Size, %:	1.28	7.5
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	September 2024	
Manufacturing Location:	Oberbruch, Heilsberg, Germany	


Quality Assurance 3/3/2025

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-252-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number 92M0105492	Serial Number 92M0105492001	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
64	50	4	1.4	FUZZBALL
203	50	2	0.7	SPLICE

Final Footage 215	Net Feet 214	Credit Method A	Defect % 0.233%	Date 5/7/2025
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Inspection Stamp 	Date 5/7/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0105492	92M0105492002	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
6	50	24	8.3	RESIN LEAN
15	50	1	0.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
135	132	A	1.543%	5/7/2025

Inspection Stamp	Date
	5/7/2025

ROLL CONDITION LOG

Batch Number 92M0105492	Serial Number 92M0105492003	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
124	50	0	0.0	FUZZBALL
147	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 5/7/2025
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Inspection Stamp 	Date 5/7/2025
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ROLL CONDITION LOG

Batch Number 92M0105492	Serial Number 92M0105492004	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 220	Net Feet 220	Credit Method A	Defect % 0.000%	Date 5/8/2025
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Inspection Stamp 	Date 5/8/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0105492	92M0105492005	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage	Net Feet	Credit Method	Defect %	Date
70	70	A	0.000%	5/7/2025

Inspection Stamp	Date
	5/7/2025